HYPOTHESIS 1: Response Time vs CSAT

Hypothesis 1 (H1):
Calls handled within SLA have a higher average CSAT score compared to calls handled above SLA, indicating that faster response times improve customer satisfaction and retention.

Hypothesis Testing

- The customer call dataset was analyzed using Pivot Tables in Excel.
- The average CSAT score was calculated for each response time category:
  Within SLA
  Below SLA
  Above SLA
- CSAT score was used for customer retention, as higher satisfaction increases the likelihood of retention.

Key Findings:

response_time	Average of csat_score
Above SLA	5.60
Below SLA	5.57
Within SLA	5.52
Grand Total	**5.54**

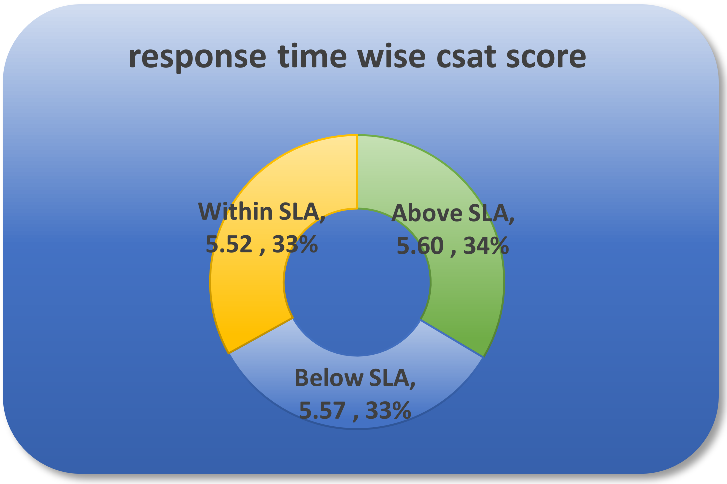
Observations:

Calls handled "Above SLA" recorded the highest average CSAT score.
Calls handled "Within SLA" had a slightly lower CSAT score.

The variation in CSAT across response time categories is minimal, indicating a weak relationship between response time and customer satisfaction.

Therefore Hypothesis 1 is rejected,

Although faster response times are operationally desirable, the analysis indicates that response time alone does not significantly improve customer satisfaction. Customers may value quality of resolution over speed of response.



Recommendations:

- Focus on Resolution Quality, Not Just Speed
- SLA compliance should be evaluated alongside CSAT and sentiment rather than as a standalone performance indicator.
- Equip agents with better tools and training to handle complex cases efficiently without compromising service quality.
- Monitor customer sentiment along with response time to better understand customer experience.

HYPOTHESIS 2: Impact of Sentiment on CSAT

Hypothesis 2 (H2):
Calls with positive or very positive sentiment have higher CSAT scores compared to calls with neutral or negative sentiment, indicating that customer sentiment significantly influences customer satisfaction and retention.

Hypothesis Testing

- Customer sentiment data was analyzed using Pivot Tables in Excel.
- Sentiment categories analyzed were:
  Very Negative
  Negative
  Neutral
  Positive
  Very Positive
- The average CSAT score was calculated for each sentiment category.
- CSAT score was used for customer retention, as higher satisfaction indicates a higher likelihood of continued customer engagement.

Key Findings:

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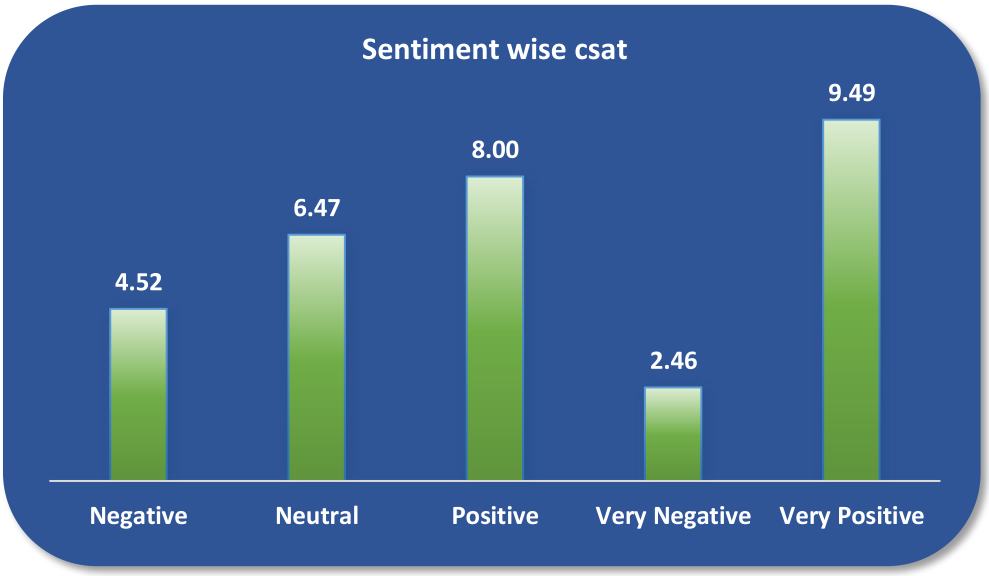
Sentiment	Average of csat_score
Negative	4.52
Neutral	6.47
Positive	8.00
Very Negative	2.46
Very Positive	9.49
Grand Total	**5.54**

Observation:

- A strong positive relationship exists between customer sentiment and CSAT score.
- CSAT scores increase consistently as sentiment improves from very negative to very positive.
- Calls with very positive sentiment recorded the highest CSAT score (9.49).
- Calls with very negative sentiment recorded the lowest CSAT score (2.46).
- The difference between extreme sentiment categories is significant, highlighting sentiment as a key driver of customer satisfaction.

Therefore Hypothesis 2 is ACCEPTED.

The analysis clearly demonstrates that customer sentiment has a strong and direct impact on CSAT scores. Positive emotional interactions result in significantly higher customer satisfaction, which directly contributes to improved customer retention.



Recommendations:

- Train customer service agents to handle interactions empathetically and professionally.
- Encourage active listening to convert negative sentiment into neutral or positive outcomes.
- Implement real-time sentiment monitoring to identify frustrated customers early in the interaction.
- Escalation for Very Negative Sentiment Cases, Route highly negative sentiment calls to experienced agents to improve resolution quality and satisfaction.
- Leverage Positive Interactions, Identify agents or call centers consistently generating positive sentiment and replicate best practices across teams.

HYPOTHESIS 3: Impact of Channel Type on CSAT

Hypothesis 3 (H3):
Digital service channels (chatbot and web) achieve higher CSAT scores compared to traditional call-center and email channels, indicating better customer experience and efficiency.

Hypothesis Testing:

- Customer interaction data was analyzed using Pivot Tables in Excel.
- The average CSAT score was calculated for each service channel:
  Call-Center
  Chatbot
  Email
  Web
- CSAT score was used as a proxy for customer retention, as higher satisfaction increases the likelihood of customer retention.

Key Findings:

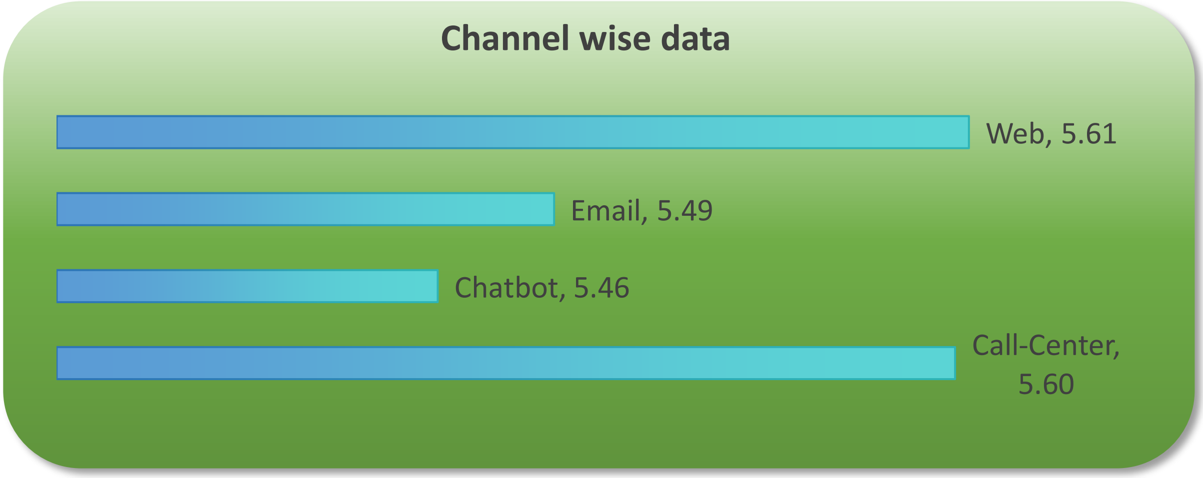
channel	Average of csat_score
Call-Center	5.60
Chatbot	5.46
Email	5.49
Web	5.61
Grand Total	**5.54**

Observations:

- CSAT scores across channels are very close, indicating a relatively consistent service experience.
- The Web channel recorded the highest average CSAT score (5.61).
- The Chatbot channel recorded the lowest average CSAT score (5.46).
- Differences across channels are minimal, suggesting channel type alone does not strongly influence customer satisfaction.

Therefore Hypothesis 3 is REJECTED.

The analysis indicates that digital channels do not significantly outperform traditional channels in terms of CSAT. Customer satisfaction appears to be influenced more by issue resolution quality and sentiment than by the communication channel used.



Recommendations:

- Ensure Consistent Service Quality Across Channels, Since CSAT scores are similar across channels, maintaining consistent service standards should be prioritized.
- Review chatbot workflows to reduce customer frustration and improve conversational accuracy.
- Use Channel Data for Cost Optimization, Even if CSAT is similar, digital channels may still reduce operational costs and should be optimized for efficiency.
- Combine Channel Strategy with Sentiment Analysis by Focus on improving sentiment within each channel rather than prioritizing one channel over another.

Overall Key Finding

Based on Analysis nearly 30000 customer service interaction for the month of October of 2020 the following key insight are found:

- Customer sentiment is the key driver of CSAT and retention.
  The CSAT score rose steadily with an increase in customer sentiment, from 2.46 for very negative to 9.49 for very positive sentiment. This shows that the emotional experience during interactions is a key factor in customer satisfaction and retention.

- Response time is not a significant factor in customer satisfaction.
  The CSAT score for calls handled above SLA was slightly higher than those handled within SLA. This shows that customers are more interested in the quality of resolution than in the time taken, especially for complex issues.

- The type of service channel has little effect on CSAT scores.
  The CSAT score for call center, web, chatbot, and email service channels was almost the same. This shows that the service experience is consistent across all channels and that customers are not satisfied with one channel over another.

- The overall CSAT level is moderate.
  The average CSAT score of 5.54 indicates that there is room for improvement in customer service effectiveness and experience.

Recommendations:

- Emphasize Customer Service Strategies Driven by Sentiment, Develop training initiatives centered on empathy, communication, and emotional intelligence. Urge customer service agents to effectively handle customers emotions, especially in negative interactions.

- Integrate Sentiment Analysis into Customer Service Operations, Apply sentiment analysis to detect high-risk customer interactions in real-time. Redirect calls with highly negative sentiments to senior customer service agents for improved handling.

- Prioritize Resolution Quality Over Speed, Although SLA is a priority, customer service agents should be encouraged to concentrate on effective issue resolution rather than on shortening call times or response times.

- Optimize Chatbots and Online Services, Enhance chatbot accuracy and dialogue effectiveness to minimize frustration. Utilize online services to efficiently process simple inquiries, allowing customer service agents to address complex issues.

- Assess Customer Service Performance Using CSAT and Sentiment Analysis, Measure customer service agent and service channel performance using a combination of CSAT, sentiment analysis, and SLA metrics instead of a single performance indicator.